

EwE linked stanza recruitment - a brief users' guide



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Compatible with Ecopath with Ecosim version 6.7 α 1
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Scope and purpose

This document provides usage instructions of linked stanza recruitment in the Ecopath with Ecosim food web modelling approach (Christensen and Walters, 2004; Heymans et al., 2016). If you have any questions, find bugs, or see ways to expand on this functionality please contact us at ewedevteam@gmail.com.

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Acknowledgements

Linked stanza recruitment was implemented by Jeroen Steenbeek after an idea by Villy Christensen under project *EISA: Ecology and management of the invasive snow crab* (<https://www.akvaplan.niva.no/en/projects-networks/eisa/>).



Summary

Occasionally, ecosystem modellers run into issues where gender-split populations need explicit modelling. This can occur when empirical data is only available for male or female specimens, or when males and females of key species have markedly different behaviours. We have made a small modification to the Ecosim with Ecosim (EwE) multi-stanza model (Christensen and Walters, 2004), that allows linking recruitment to ensure proportional spawning between two multi-stanza groups, and thus connecting two gender-split populations at their foundation.

Computations

At the first Ecosim and Ecospace time steps, a base scalar is calculated:

$$a_i = \frac{N_{Age1_i}}{N_{Age1_j}} \quad \text{Eq. 1}$$

where a is the scalar, i is a multi-stanza group whose recruitment is driven by multi-stanza group j , and N_{Age1} is the number of individuals at age 1, e.g., the number of recruits, for multi-stanza groups i and j . This base scalar serves to maintain the ratio between the number of gender-split recruits during the simulations.

For every consecutive time step, the number of multi-stanza group i recruits is then set as follows:

$$N_{Age1_i} = a_i \times N_{Age1_j} \quad \text{Eq. 2}$$

Changes to the EwE computations were minimal; the most challenging code changes were made in the Ecospace IBM model (Walters et al., 2010), where parallel computations was reorganized to ensure that multi-stanza recruitment is calculated in the correct order. Ecospace does not yet calculate recruitment success from spatial overlap of the gender-split stanza; this could be a future refinement if needed.

An example of the impact of linked recruitment is shown in Figure 1. Here, Ecosim is executed for a model that contains separate multi-stanza groups for male and female snow crab *Chionoecetes opilio*. Both multi-stanza groups consist of two life stages each: juveniles and adults. Only the male snow crabs are fished. The model contains no change in environment and fishing other than gradually intensifying effort of the snow crab fishery. Figure 1 shows three simulations: in the left panel, multi-stanza recruitment is not linked. In the middle panel, female recruitment is obtained from the male multi-stanza group. In the right panel, male recruitment is obtained from the female multi-stanza group.

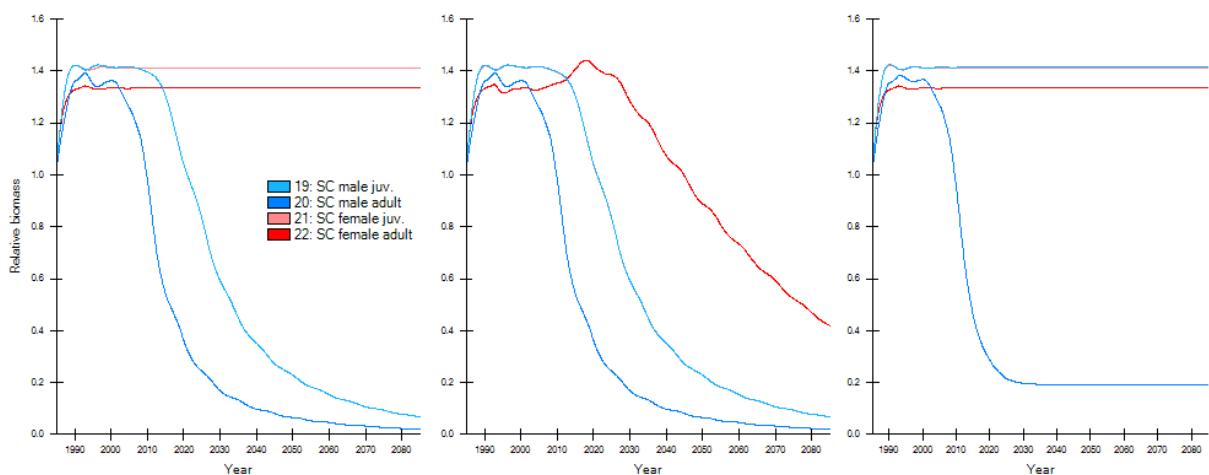


Figure 1 - Three Ecosim simulations with different recruitment linkage configurations for snow crab: no linked recruitment (Left), female recruitment linked to the fished males (Center), and fished male recruitment linked to non-fished females (Right). Note that both juvenile groups are identically parameterized in this specific case, and therefore their biomass estimates exactly overlap in the Center and Right plots.

Configuring linked stanza recruitment

First, define the male and female multi-stanza configurations in the “Define groups” interface (Figure 2).

Group name	Color	Consumer	Producer	Detritus	Part of production from primary	Multi-stanza group name	Stz (in)
12 Mudocok		☒	☐	☐	0.000		
13 Plank_fish		☒	☐	☐	0.000		
14 Plank_pisc_fish		☒	☐	☐	0.000		
15 Pisc_fish		☒	☐	☐	0.000		
16 Small_benthivore_fish		☒	☐	☐	0.000		
17 Large_benthivore_fish		☒	☐	☐	0.000		
18 Shrimp		☒	☐	☐	0.000		
19 SC male juv.		☒	☐	☐	0.000	SC male	0
20 SC male adult		☒	☐	☐	0.000	SC male	96
21 SC female juv.		☒	☐	☐	0.000	SC female	0
22 SC female adult		☒	☐	☐	0.000	SC female	96
23 Red_king_crab		☒	☐	☐	0.000		
24 Large crabs		☒	☐	☐	0.000		
25 Pred_iver		☒	☐	☐	0.000		
26 Depos_polychaete		☒	☐	☐	0.000		
27 Depos_echinoderm		☒	☐	☐	0.000		
28 Depos_other		☒	☐	☐	0.000		

Figure 2 - Define gender-split multi-stanza groups

Second, configure the two gender-split multi-stanza groups in the “Edit multi-stanza” interface (not shown here).

Third, link recruitment in the “Edit multi-stanza” interface. In the figure below, female snow crab recruitment is linked to the males. A multi-stanza group whose recruitment is linked will have its recruitment calculations overwritten, and therefore the “Recruitment Power” parameter becomes inactive (Figure 3).

Stanza group: 3: SC female

Curvature parameter, K (annual), from VBGF: 0.140

Recruitment power: 1.000

Recruitment stanza: 2: SC male

Relative biomass accumulation rate (BA/B): 0.170

Wmaturity/Winf: 0.0900

☐ Fixed fecundity

Ecosim

Forcing function for hatchery stocking:

Ecospace

☒ Recruit where spawned in Ecospace

Number at age 0 (n/km2/year): 0.00423

Normalized values vs Age graph:

Group name	Age, start (months)	Leading biomass	Biomass (t/km²)	Tot. mort. (/year)	Leading Q/B	Consumption / biomass (/year)	Spawning (proportion)
21 SC female juv.	0	☐	0.000379	0.230	☐	3.383	0.000
22 SC female adult	96	☒	0.000112	0.450	☒	2.005	1.000

Figure 3 - Linking the recruitment of a multi-stanza configuration to another multi-stanza.

References

- Christensen, V., Walters, C.J., 2004. Ecopath with Ecosim: methods, capabilities and limitations. *Ecol. Model.* 172, 109–139. <https://doi.org/10.1016/j.ecolmodel.2003.09.003>
- Heymans, J.J., Coll, M., Link, J.S., Mackinson, S., Steenbeek, J., Christensen, V., 2016. Best practice in Ecopath with Ecosim food-web models for ecosystem-based management. *Ecol. Model.* 331, 173–184. <https://doi.org/10.1016/j.ecolmodel.2015.12.007>
- Walters, C.J., Christensen, V., Walters, W., Rose, K.A., 2010. Representation of multistanza life histories in Ecospace models for spatial organization of ecosystem trophic interaction patterns. *Bull. Mar. Sci.* 86, 439–459.